

Calculus (PR12)

Week 2: Differentiation Techniques

Announcement

Please submit your preference for the assignment before 1 O'clock today.

Learning Goals



Differentiation rules

- Use the **product and quotient rules**
- Apply the **chain rule**



Trigonometric and implicit functions

- Compute derivatives of **trigonometric functions**
- Use **implicit differentiation** when variables are **interconnected**



Advanced differentiation

- Compute and interpret **higher-order derivatives**
- Connect differentiation techniques to **solve complex problems**

Differentiation and AI

Question:

Which AI application do you think uses derivatives MOST?

1. TikTok recommendations
2. Self-driving cars
3. Chatbots
4. Face recognition

Relevance: Neural network training, Optimization, Computer vision. Robotics

Quick Recap

Review:

- Power rule
- Sum rule

Example:

$$\frac{d}{dx} (x^3 + 2x) = ?$$

Question: Could we differentiate $x^2 \sin x$ using only these rules?

Product Rule

Example:

$$f(x) = x^2 (x^3 + 1)$$

Product Rule Formula:

$$\frac{d}{dx} [f(x)g(x)] = \dot{f}(x)g(x) + f(x)\dot{g}(x)$$

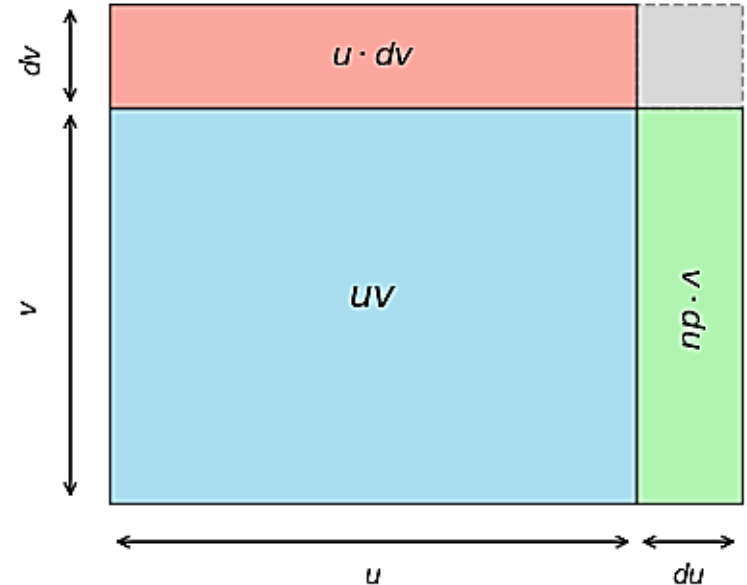
Worked Example

Example:

$$f(x) = x^2 (x^3 + 1)$$

Solution:

$$\hat{f}(x) = 2x(x^3 + 1) + x^2(3x^2)$$



Question: Why are there TWO terms in the answer?

Quotient Rule

Example:

$$f(x) = \frac{x^2 + 1}{x}$$

Product Rule Formula (*"Low d-high minus high d-low."*) :

$$\frac{d}{dx} \frac{f(x)}{g(x)} = \frac{\dot{f}(x)g(x) - f(x)\dot{g}(x)}{[g(x)]^2}$$

Question: Why can't we divide derivatives?

Worked Example

Example:

$$f(x) = \frac{x^2 + 1}{x}$$

Solution:

$$\hat{f}(x) = \frac{(2x)(x) - (x^2 + 1)(1)}{x^2}$$

Spot the Mistake

Example (*Incorrect work*):

$$\frac{d}{dx} \left(\frac{x^2 + 1}{x} \right) = \frac{2x}{1}$$

Question: What went wrong?

Chain Rule

Example :

$$f(x) = (3x + 1)^5$$

Question:

- **What is the outer function?**
- **What is inside?**

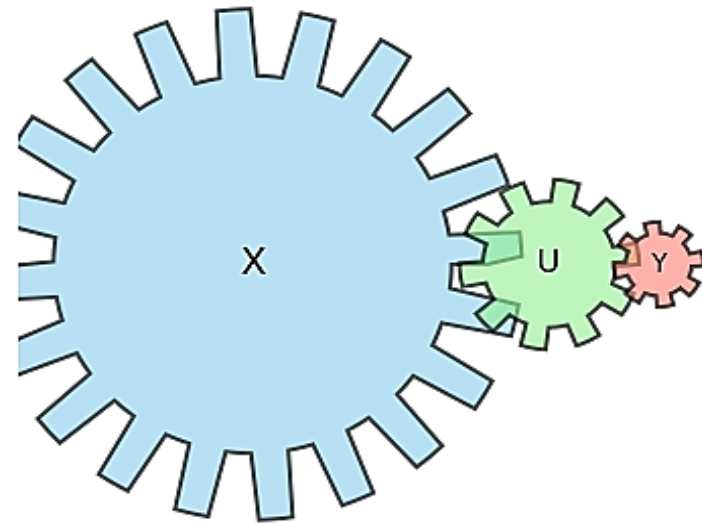
Chain Rule Formula

$$\frac{d}{dx} [f(g(x))] = f'(g(x)) \cdot g'(x)$$

Key point: Differentiate outside first.

In another word:

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$



Worked Example

Example:

$$f(x) = (3x + 1)^5$$

Solution:

$$\hat{f}(x) = 15(3x + 1)^4$$

Multi-Layer Chain Rule

Example:

$$f(x) = (x^2 + 3x + 1)^4$$

$$\hat{f}(x) = 4(x^2 + 3x + 1)^3 (2x + 3)$$

Rule Classification

Which rule?

Case 1:

$$f(x) = e^x x^2$$

Case 2:

$$f(x) = \sin(4x)$$

Case 3:

$$f(x) = \cos x / x$$

Core Trigonometric Derivatives

$$\frac{d}{dx}(\sin x) = \cos x$$

$$\frac{d}{dx}(\cos x) = -\sin x$$

Also:

$$\frac{d}{dx}(\tan x) = \sec^2 x$$



Worked Trigonometric Example

$$f(x) = (3 \sin x - 2 \cos x)$$

$$\frac{d}{dx} f(x) = 3 \cos x + 2 \sin x$$

Worked Trigonometric Example

$$f(x) = \sin(4x)$$

Which answer is correct?

Option 1:

$$\cos(4x)$$

Option 2:

$$4 \cos(4x)$$

Option 3:

$$-4 \sin(4x)$$



Implicit Functions

Example:

$$x^2 + y^2 = 25$$

An **implicit function** is a relationship where variables are mixed in one equation instead of one being solved clearly in terms of the other.

Implicit Differentiation Steps

Differentiate both sides:

$$x^2 + y^2 = 25$$

$$2x + 2y \frac{dy}{dx} = 0$$

Then solve for $\frac{dy}{dx}$:

$$\frac{dy}{dx} = -\frac{x}{y}$$

Higher Derivatives - Second Derivatives

Example:

$$f(x) = x^3 + 3x^2$$

First derivative:

$$3x^2 + 6x$$

Second derivative:

$$6x + 6$$

Check?

What feels clear now?
What still feels uncertain?

Remember:
To practice, practice and practice!

Next week:
Optimization, Taylor and Maclaurin series.

?

Questions

- Follow-up reading: Calculus: A Complete Course, (sections 2.3 – 2.6 & 2.9)
- Joining the Discussion Groups and the Lab

